

Synopsis

on

Campus Grievance Redressal System

MCA (Integrated) – IV Semester

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1. Introduction

Educational institutions frequently face various campus-related issues such as malfunctioning electrical equipment, faulty computer systems, damaged furniture, non-working projectors, and inadequate water supply. Problems related to classroom furniture like broken chairs, damaged tables, and benches also affect the learning environment and student comfort.

Currently, most of these issues are reported verbally or through informal communication, which often results in delayed resolution, miscommunication, and lack of accountability.

The Campus Grievance Redressal System is a web-based application designed to provide a centralized digital platform for reporting, tracking, and resolving campus-related grievances. The system allows students, faculty, and staff to submit complaints online along with image proof, ensuring transparency, faster resolution, and effective monitoring by the administration.

2. Problem Statement

There is no centralized and structured digital system to manage campus grievances related to infrastructure, furniture, and equipment. Complaints are currently communicated through verbal conversations or emails, leading to poor tracking, delayed responses, and no proper evidence of the reported problem.

Additionally, the absence of image-based proof makes it difficult for administrators to assess the severity and authenticity of complaints. This lack of transparency negatively impacts campus maintenance and overall institutional efficiency.

3. Aim & Objectives

Aim: To develop a secure, efficient, and user-friendly web-based Campus Grievance Redressal System that enables users to report campus issues with image evidence and allows administrators to resolve them effectively.

Objectives:

- To provide a centralized platform for submitting campus grievances
- To allow users to report issues related to infrastructure, furniture, and equipment
- To enable image upload functionality as proof of the reported issue
- To categorize complaints based on type and location
- To provide administrators with a dashboard for complaint management
- To ensure secure access using authentication and authorization
- To notify users about complaint status through email alerts

4. Scope of Project

The scope of the project includes handling grievances related to:

- Electrical issues (lights, fans, switches, wiring)
- Computer hardware issues (CPU, keyboard, monitor, mouse)
- Projectors and classroom equipment
- Furniture issues (broken chairs, damaged tables, benches)
- Library and faculty room infrastructure
- Water supply and basic campus facilities
- Image upload feature for providing visual proof of complaints

5. Literature Review

Various studies on digital grievance management systems emphasize the importance of automation, transparency, and evidence-based complaint handling. Existing systems used in educational institutions mainly support basic complaint registration but lack advanced features such as image uploads, secure authentication, and real-time status tracking.

The proposed system enhances traditional grievance management solutions by incorporating secure login, structured categorization, image-based proof, and automated notifications, making it more efficient and reliable

6. Project Schedule / Timeline

Phase	Description	Duration
Requirement Analysis	Understanding system requirements	1 week
Understanding system requirements	Database & architecture design	2-3 weeks
Backend Development	Spring Boot & Hibernate implementation	4-6 weeks
Frontend Development	Thymeleaf UI development	7 weeks
Security Integration	JWT & Spring Security	8 weeks
Testing & Debugging	Functional & security testing	9-10 weeks

7. Proposed System / Methodology

The proposed system follows a layered architecture:

- Users register and log in securely
- Complaints are submitted through an online form
- Users can upload images as proof of the issue
- Complaint data is stored in a MySQL database using Hibernate (JPA)
- Administrators access complaints through a dashboard
- Complaint status is updated (Pending, In-Progress, Resolved)
- Email notifications are sent to users regarding status updates

The system uses Spring Boot for backend services, Thymeleaf for frontend rendering, and JWT-based authentication for secure access.

8. Tools and Technologies Used

Backend: Java, Spring Boot, Hibernate (JPA)

Security: Spring Security with JWT

Database: MySQL

Frontend: Thymeleaf

File Upload: Multipart File Handling

Notifications: Email API

Tools: Maven, Postman, Git

9. Expected Outcomes

- Centralized and transparent grievance management
- Faster resolution of campus infrastructure issues
- Image-based proof for accurate complaint verification
- Secure and role-based access control
- Improved campus facility maintenance
- Enhanced student and faculty satisfaction

Signature of Student

Signature of Guide